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Date: 07/27 – 07/31

Weekly Report

Week: 07/27 – 07/31

Subject of This Week:

- **Traveling Politician Project** (In Progress)
 - Set up project structure and loaded state capitals data with coordinates.
 - Implemented Haversine formula for distance calculation.
 - Built greedy algorithm to find a route starting from Des Moines, Iowa, visiting all state capitals, and ending in Washington, DC.
 - Currently testing the route output and preparing for optimization.
 - Plan to add 2-opt improvement and route visualization.
- **NASPO Catalog Project** (Improvements)
 - Split single search box into two separate fields: Vendor search and Product/Description search.
 - Updated price display format to show commas and two decimal places (e.g., \$6,000.00).
 - Used Django humanize template filters for better formatting.
- **MMIS AI – Electronic Claim Entry & Submission Portal**
 - Downloaded and ran Chao's PATO application locally.
 - Reviewed the database schema and C4 model diagrams.

Notes:

- Haversine formula calculates great-circle distance between two points on Earth using latitude and longitude.
- The greedy algorithm selects the nearest unvisited capital at each step.
- 2-opt optimization will improve the route by swapping path segments to reduce total distance.
- Django humanize filters provide human-readable number formatting without custom code.

Plan for Next Week:

- Complete Traveling Politician Project with 2-opt optimization and route visualization.
- Continue MMIS AI Project development.
- Continue MIT Python lectures.

Hours Worked: 18

Intern Checklist:

1. Receive the internship offer letter
2. Accept the terms of the offer letter
 - 2.1. Create a Telegram account and be added into group chat
 - 2.2. Create a LinkedIn account (Connect with Verbus, and other interns in the program. Then join MMIS Professionals and Mostly Medicaid (Medicaid Professionals))
 - 2.3. Create a GitHub account
 - 2.4. Please use Mermaid in your documentation.
3. Recommended AI assistant is CLAUDE
4. Attend the daily standup calls
5. Take notes and produce weekly report (due Friday's, Saturdays at the latest)
 - 5.1. PDF (EMRTS_Weekly_report_date_firstname_lastname.pdf)
 - 5.2. JSON (EMRTS_Weekly_report_date_firstname_lastname.json)
6. Watch the Python Lectures from MIT (Lectures 1-16)
 - 6.1. <https://ocw.mit.edu/courses/6-0001-introduction-to-computer-science-and-programming-in-python-fall-2016/>
10. Work on the Assigned Project
 - 10.1. State Capitals Project - completed
 - 10.2. 4-Gallon Water Bucket Problem - completed
 - 10.3. TABIND Scrabble Project - completed
 - 10.4. FizzBuzz Project - completed
 - 10.5. Provider Lookup Project - completed
 - 10.6. Bookstore Project - completed
 - 10.7. Provider Exclusion List Project - completed
 - 10.8. NASPO Catalog Project - completed
 - 10.9. MMIS AI Project - in progress
 - 10.10. Traveling Politician Project - in progress

EMRTS

<https://emrts.us/>

GitHub

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